

THARUN DERANGULA

Software Engineer | Applied AI & Full-Stack | Distributed Systems

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SUMMARY

Software engineer with five years of experience building full-stack applications and distributed systems using Python, Java, and TypeScript. Currently developing LLM and RAG features for Walmart's Sparky shopping assistant, including vector retrieval, evaluation pipelines, and cart services within commerce systems handling 20M+ daily events.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript (ES6+), Go, SQL

Frontend: React, Next.js (SSR), Redux, Tailwind CSS, Jest, React Testing Library, Cypress

Backend: FastAPI, Spring Boot, Node.js, Express.js, REST APIs, GraphQL, gRPC, Microservices, Kafka, Redis, OpenAI API, Anthropic API

AI/ML: Generative AI, LLM Integration & Evaluation, RAG, Embeddings, pgvector, Vector Search, LangChain, AI Agents, Prompt Engineering

Cloud & DevOps: GCP, AWS, Docker, Kubernetes, Terraform, Helm, GitHub Actions, CI/CD, Datadog, Prometheus, Grafana

Databases: PostgreSQL, MongoDB, Redis, Snowflake, Elasticsearch, BigQuery, DynamoDB

Testing & Quality: pytest, JUnit, Jest, React Testing Library, Cypress, TDD, A/B Testing

Practices & Tools: System Design, Event-Driven Architecture, MLOps, Agentic Workflows, Agile/Scrum, Gemini, Claude Code, GitHub Copilot, Codex

EXPERIENCE

Software Engineer, Walmart Global Tech — Sunnyvale, CA

April 2025 – Present

- Shipped conversational shopping and cart features for Sparky, Walmart's AI shopping agent, across web and mobile using **React**, **Next.js**, and **TypeScript**, connecting the UI to **Python** and **Java** services and Walmart's Wallaby **LLM**.
- Improved recommendation relevance by 20% in A/B tests by building **Python RAG** evaluation pipelines, tuning **pgvector** retrieval, and integrating Walmart's Wallaby **LLM** through the Element ML Platform.
- Built **Python** and **FastAPI** services for prompt orchestration, embedding generation, and retrieval evaluation, adding **pytest** regression suites to catch **RAG** failures and **GitHub Actions** quality gates enforcing 85% minimum service test coverage.
- Developed intent-routing and cart auto-build services using **Java**, **Spring Boot**, **Kafka**, and **PostgreSQL**, connecting conversational shopping requests to backend cart workflows within systems processing 20M+ daily shopping events.
- Reduced recommendation API p95 latency by 35% through **Redis** caching, connection pooling, and **PostgreSQL** query and index tuning, maintaining sub-50ms median latency at 5K+ requests/sec during holiday peaks.
- Secured shopping and cart APIs with **OAuth 2.0**, **JWT**, and role-based access control, enforcing least-privilege access to customer cart and account data across web and mobile clients.
- Halved deployment time from approximately 20 to 10 minutes and enabled zero-downtime releases using **Docker**, **Kubernetes**, **Terraform**, and **GitHub Actions** on **WCNP**, with **Datadog** and **Grafana** monitoring.
- Authored technical design documents and led architecture reviews for cart automation, intent routing, and **RAG** evaluation services, resolving integration decisions with product, ML, and platform teams before implementation.
- Adopted **Claude Code**, **GitHub Copilot**, **OpenAI Codex**, and **Gemini** for refactoring and test generation, reducing PR turnaround by roughly one-third and doubling test-writing speed while requiring CI checks before merging.

Software Engineer, Kennesaw State University — Kennesaw, GA

January 2024 – December 2024

- Built KSU's Campus Hub using **React**, **Next.js**, and **TypeScript**, bringing Owl Express, D2L Brightspace, DegreeWorks, and KSUMail together in a responsive platform serving 40,000+ students and faculty.
- Integrated university single sign-on into Campus Hub and implemented role-based access controls with protected API routes, enabling students and faculty to securely access personalized academic information.
- Improved dashboard load time by 30% by developing **Node.js** and **Spring Boot REST APIs** for Ellucian Banner and DegreeWorks integrations, streamlining data retrieval for student-facing campus dashboards.
- Reduced manual data correction by 40% with **Java** and **Spring Boot** services on **PostgreSQL** that synchronized enrollment, grades, and financial aid records through **REST APIs** and webhooks.
- Halved deployment cycle time using **Docker** and **GitHub Actions** for Campus Hub services on **AWS EC2**, with static assets on **S3**, supporting releases during peak registration periods.
- Maintained 80%+ test coverage using **Jest** and **React Testing Library** to catch UI regressions earlier, collaborating with UITs teams through Agile sprints, technical design reviews, and code reviews.

Software Engineer, HCL Tech — India

May 2020 – December 2022

- Built invoice tracking and payment workflows for enterprise banking customers across regions using **React** and **TypeScript** frontends backed by **Java** and **Spring Boot** APIs, supporting daily financial operations.
- Improved backend responsiveness during peak banking activity by optimizing connection pooling across **Java** and **Spring Boot** services, supporting invoice tracking and payment workflows used by enterprise customers.
- Developed real-time transaction pipelines with **Kafka**, implementing idempotent consumers and dead-letter queues to handle duplicate events and isolate processing failures during high-volume settlement windows across banking workflows.
- Cut batch processing time by 45% by building ETL pipelines with **Python**, **pandas**, **SQL**, and **GCP BigQuery**, loading transaction data into partitioned warehouse tables for finance and compliance teams.
- Reduced data quality incidents by 35% through automated validation and reconciliation checks in **Airflow** workflows, improving the consistency of transaction datasets used for finance and compliance reporting.
- Modernized legacy banking backends into microservices using **Docker**, **Terraform**, **GCP Cloud Run**, and **GKE**, reducing production defects by 40% through **TDD**, **JUnit**, **pytest**, and automated **GitHub Actions** checks.

EDUCATION

Master of Science, Information Technology | Kennesaw State University

Bachelor of Science, Computer Science & Engineering | Lovely Professional University

Certifications: Google Professional Cloud Architect | AWS Certified Solutions Architect | Microsoft Generative AI Engineering Professional